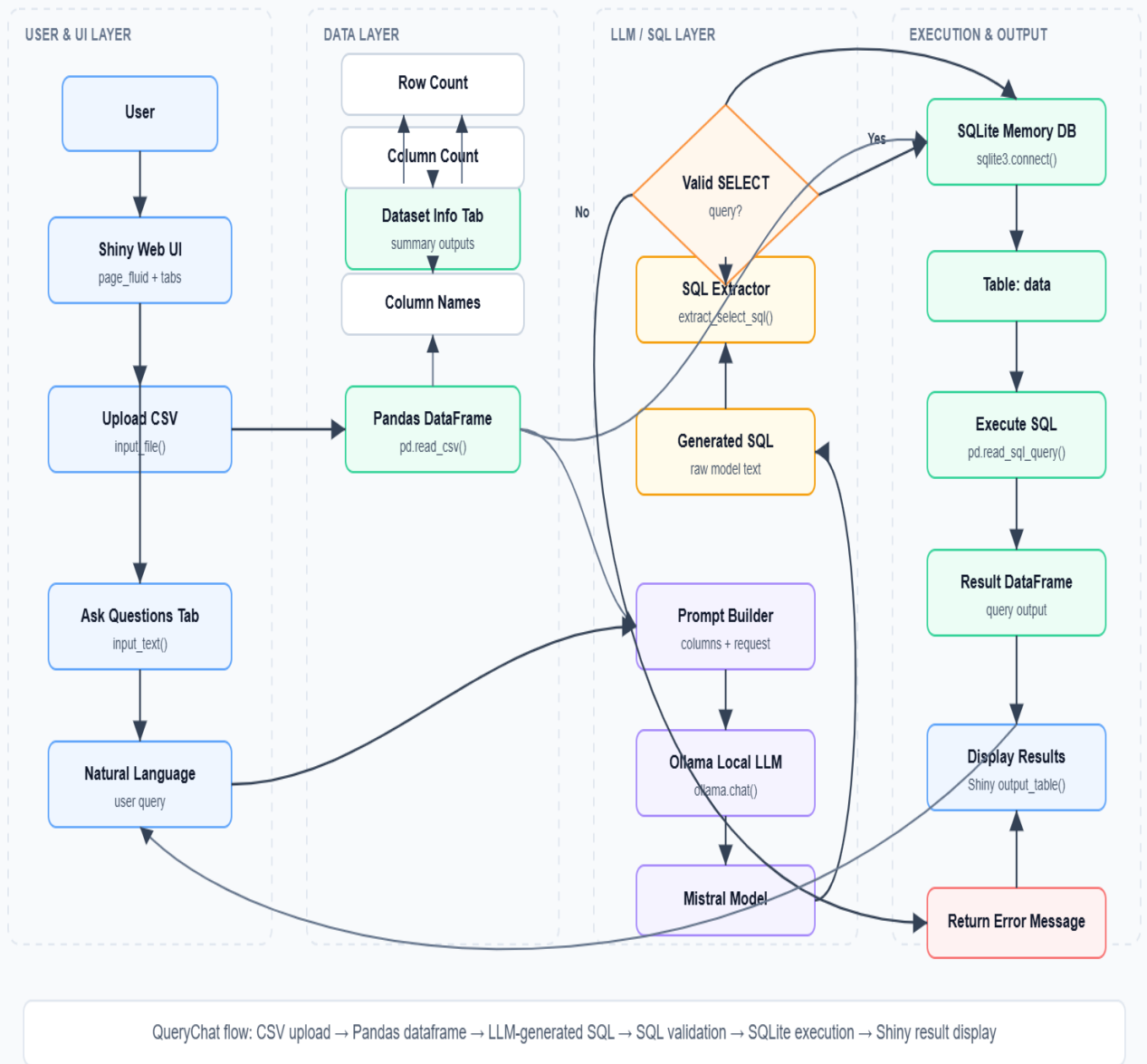


QueryChat Orchestration

Ask questions about CSV data using Shiny + Pandas + Ollama/Mistral + SQLite



CSV Upload

↓

Pandas DataFrame

↓

Dataset Summary + Column Info

↓

User Question

↓

Ollama / Mistral generates SQL

↓

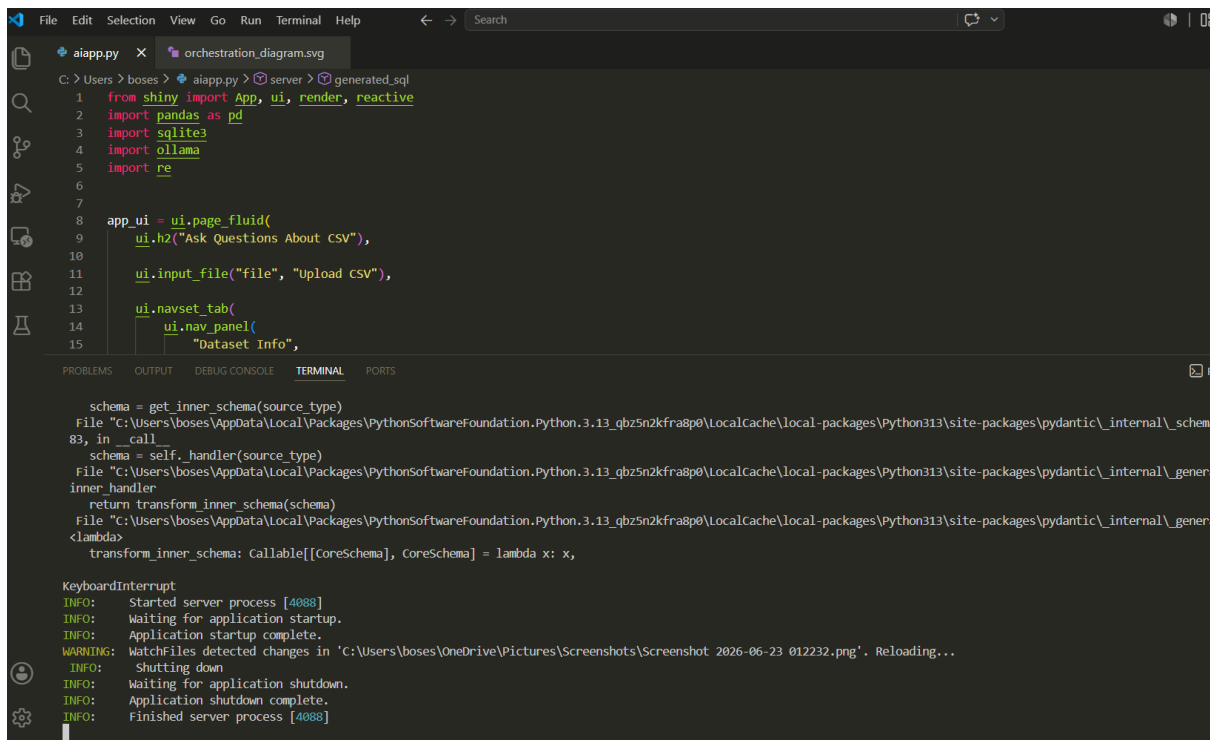
SQL is cleaned and validated

↓

SQLite runs query on uploaded data

↓

Results shown in Shiny UI



The image shows a VS Code editor window with a Python script named `aiapp.py` and its terminal output. The script is a Shiny app that uses `shiny`, `pandas`, `sqlite3`, `ollama`, and `re` modules. It defines a `ui` and a `server` function. The `ui` function creates a page with a title "Ask Questions About CSV", an input file, and a tab for "Dataset Info". The `server` function handles the input file, generates SQL using `ollama`, and returns the results.

```
1 from shiny import App, ui, render, reactive
2 import pandas as pd
3 import sqlite3
4 import ollama
5 import re
6
7
8 app_ui = ui.page_fluid(
9     ui.h2("Ask Questions About CSV"),
10
11     ui.input_file("file", "Upload CSV"),
12
13     ui.navset_tab(
14         ui.nav_panel(
15             "Dataset Info",
```

The terminal output shows the execution of the script. It starts with a `KeyboardInterrupt` message, followed by `INFO: Started server process [4088]`. The application starts up successfully, but then a `WARNING: WatchFiles detected changes in 'C:\Users\bores\OneDrive\Pictures\Screenshots\Screenshot 2026-06-23 012232.png'. Reloading...` message is shown. The application then shuts down, with `INFO: Shutting down`, `INFO: Waiting for application shutdown.`, `INFO: Application shutdown complete.`, and finally `INFO: Finished server process [4088]`.

Ask Questions About CSV

Upload CSV

Browse...	No file selected
-----------	------------------

Dataset Info [Ask Questions](#)

Basic Dataset Information

Information	Value
Status	No CSV uploaded yet

Column Names

Ask Questions About CSV

Upload CSV

Browse...	clinical_lab_dataset.csv
Upload complete	

Dataset Info	Ask Questions
--------------	-------------------------------

Basic Dataset Information

Information	Value
Number of rows	200
Number of columns	7

Column Names

Column Number	Column Name
1	PatientID
2	Age
3	Sex
4	Hemoglobin
5	Platelets
6	ALT
7	AdverseEvent

Ask Questions About CSV

Upload CSV

Browse...	clinical_lab_dataset.csv
Upload complete	

[Dataset Info](#)

[Ask Questions](#)

Ask a question:

how many observations are present in 1

Generated SQL

```
SELECT COUNT(*) FROM data
```

Results

COUNT(*)

200

Ask Questions About CSV

Upload CSV

Browse...

clinical_lab_dataset.csv

Upload complete

Dataset Info

Ask Questions

Ask a question:

Create a Frequency table of Adversever

Generated SQL

```
SELECT `Sex`, COUNT(`AdverseEvent`) AS Frequency
FROM data
GROUP BY `Sex`
```

Results

Sex	Frequency
Female	103
Male	97

Ask Questions About CSV

Upload CSV

Browse...

clinical_lab_dataset.csv

Upload complete

Dataset Info

Ask Questions

Ask a question:

How many patients present with Sex M.

Generated SQL

```
SELECT COUNT(PatientID) FROM data WHERE Sex = 'Male'
```

Results

COUNT(PatientID)
97

Ask Questions About CSV

Upload CSV

Browse...	clinical_lab_dataset.cs
Upload complete	

Dataset Info

Ask Questions

Ask a question:

Min, max and mean values of ALT

Generated SQL

```
SELECT MIN("ALT"), MAX("ALT"), AVG("ALT") FROM data
```

Results

MIN("ALT")	MAX("ALT")	AVG("ALT")
3.031134	55.733598	29.627294

Ask Questions About CSV

Upload CSV

Browse...

clinical_lab_dataset.csv

Upload complete

Dataset Info Ask Questions

Ask a question:

patients with Age less than 50

Generated SQL

```
SELECT * FROM data WHERE Age < 50
```

Results

PatientID	Age	Sex	Hemoglobin	Platelets	ALT	AdverseEvent
3	46	Male	16.298662	249.074343	25.123938	0
4	32	Male	14.210749	235.567068	25.674418	1
6	25	Male	14.484830	208.638453	25.790155	0
8	38	Male	14.680627	326.636946	50.754008	0
11	36	Female	14.945064	284.507200	42.012139	0
12	40	Male	14.119171	229.938976	25.919246	0
13	28	Male	14.733090	261.204624	9.618755	0
14	28	Male	16.345189	250.629620	19.919137	1
15	41	Female	13.131918	254.883805	11.292081	0
19	41	Male	13.384347	322.557180	33.269274	0
20	20	Female	14.011728	297.963541	27.808995	0
21	39	Male	13.915036	357.659123	38.294056	0